AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 6 · STUDENT LESSON

Label Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 8: operational label definitions reduce avoidable disagreement.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Write an observable label rule with boundaries and a review option. Try the recall before opening notes.

Your learning target

I can write an observable **label rule** with boundaries and a review option.

I can measure **agreement** using matched items and an explicit denominator.

I can investigate dis**agreement** and record justified label decisions.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Ambiguous definitions and inconsistent **annotation** place an upper bound on learnable performance.



Your target

LEARNING TARGET Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

Builder name and experiment date:

Readiness for the connected explanation

Why should an unknown duration remain different from zero in a dataset card?

This week’s end goal

By the end, I can explain label quality using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

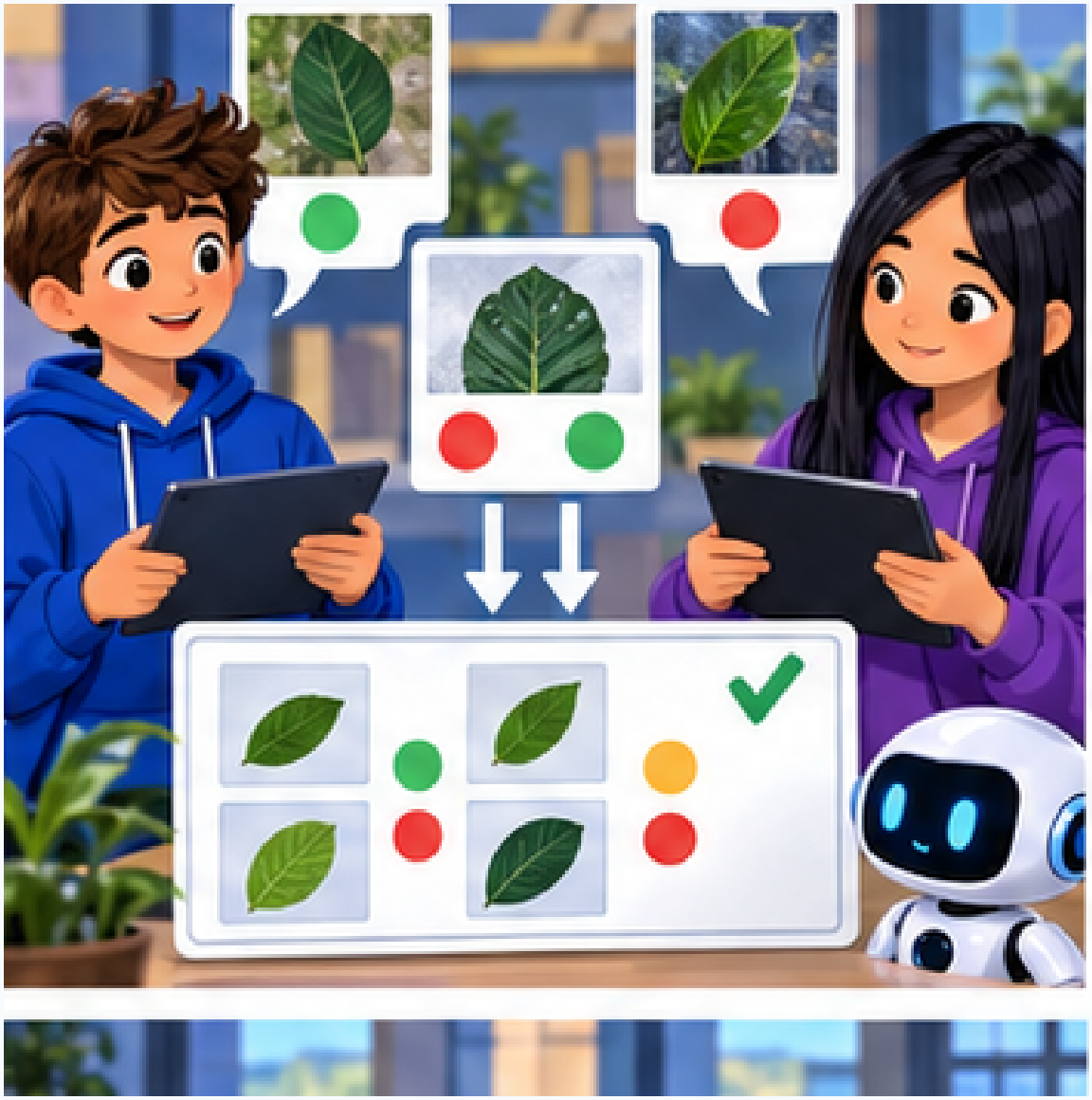
Notice the evidence

> Investigate the supplied Label Quality case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use label quality to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A model learns from examples with recorded targets. If two people interpret a target differently, their labels can mix different tasks.

This week you will write a **label rule**, compare independent labels and keep a record of disputed cases. We use invented corridor descriptions, not real images or a system for controlling a robot.

The classroom task is to label whether a marked practice lane is blocked in a supplied description.

Rule v1 says blocked means an object touches or crosses either lane boundary or lies inside the lane; clear means all objects are outside both boundaries and the entire lane is visible.

Use review if visibility or position is insufficient to decide. Review is a workflow status for a person to inspect, not a third physical condition of the lane.

Description	Rule v1 result	Reason
A box lies inside a fully visible lane	blocked	Inside satisfies the rule
A wheel touches the boundary line	blocked	Touching is explicitly included
All objects are outside and the lane is fully visible	clear	Both clear conditions hold
A curtain hides part of the lane	review	Hidden space cannot be checked

These are exercise definitions, not real robot safety rules. Do not infer that a lane is physically safe from this label. The label describes only the supplied **evidence** under this version of the rule.

STUDENT LESSON · 3 OF 18

See how it works

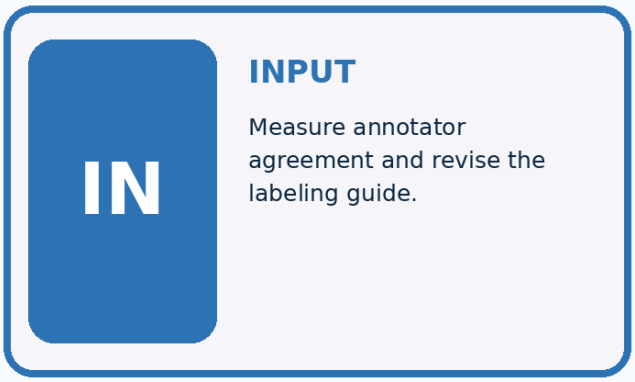
> Follow the mechanism for Label Quality. Connect each labelled input, deciding step and

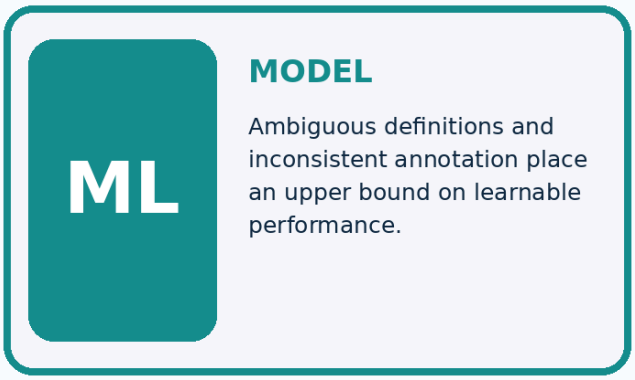
> result.

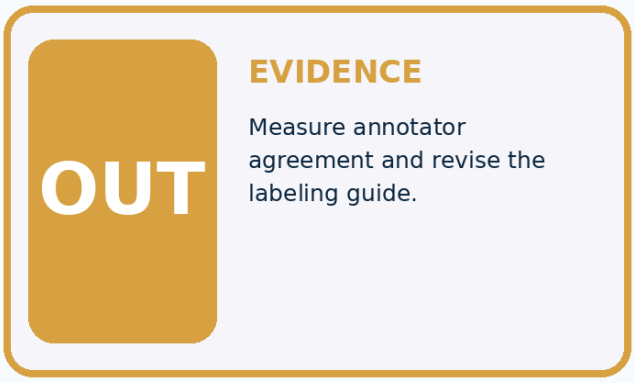
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Measure **annotator agreement** and revise the labeling guide.

MODEL IDEA Ambiguous definitions and inconsistent **annotation** place an upper bound on learnable performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Two **annotator**s label 20 leaves and agree on 16. Observed **agreement** = 16 ÷ 20 = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Two **annotator**s used an earlier vague instruction, mark an obstructed lane, on four invented items. Their labels are shown in the code. A and D match; B and C differ.

Compare the same item IDs, not merely the first row in two differently ordered files.

```
pairs = [("A", "blocked", "blocked"),
         ("B", "clear", "blocked"),
         ("C", "review", "clear"),
```

```
          ("D", "clear", "clear")]
agree = 0
for item_id, first, second in pairs:
    if first == second:
        agree = agree + 1
print(agree, len(pairs))
```

The **output** is 2 4: two matching labels among four paired items, so observed exact **agreement** is 2/4 or 50 percent. Here review is included as a label value in the comparison: review versus clear is a dis**agreement**.

This simple count is not model accuracy and does not establish which labels are correct.

Follow a disagreement back to evidence

OBSERVE B has two labels	INVESTIGATE Read boundary rule	RECORD Keep labels and reason
OBSERVE C includes review	INVESTIGATE Check visible evidence	RECORD Retain uncertainty if needed
OBSERVE A and D match	INVESTIGATE Check rule as well	RECORD Agreement is not proof

Preserve independent labels before discussion. Shared mistakes can produce agreement, and missing evidence can prevent a justified final label.

The code assumes one valid pair per unique item ID and allowed label strings. It does not validate IDs or spelling. Use a nonempty paired set before calculating a fraction; zero paired items gives no **agreement** estimate.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

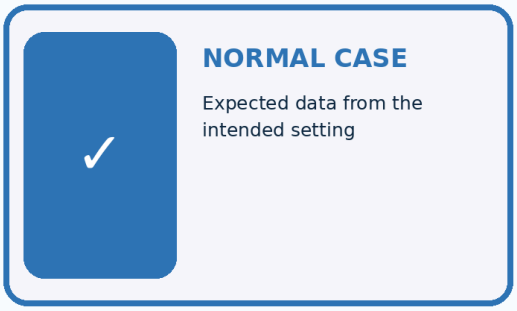
Compare the cases

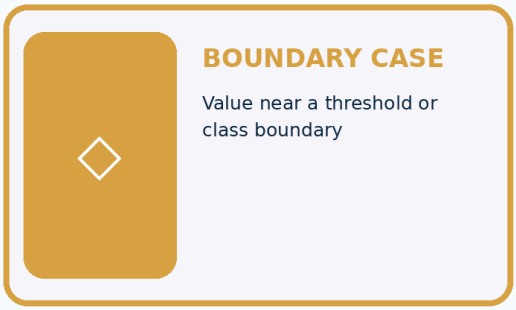
> Compare the supplied cases for Label Quality. Identify the detail that changes the conclusion.

EVALUATION DESIGN

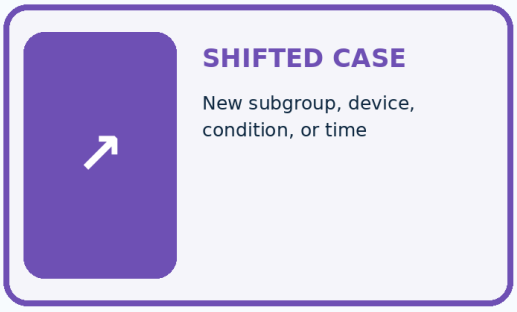
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Z receives a new fully visible image with no brown spots.

Predicted result

Prediction: Label the new **observation** healthy under the toy rule.

Why the result follows

Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Label Quality to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
ANNOTATION
human assignment of a label
2
LABEL RULE
observable criterion for assigning a class
3
AGREEMENT
fraction of labels that match
4
ADJUDICATION
resolve disagreements using evidence

Term	Precise meaning	My ML example
annotation	human assignment of a label	
label rule	observable criterion for assigning a class	
agreement	fraction of labels that match	
adjudication	resolve disagreements using evidence	

MISCONCEPTION TO REPAIR The model can learn the correct concept even when people label the same example differently.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Label rule	A written definition for assigning an outcome to an example
Annotator	A person who assigns labels using a stated rule
Agreement	How often annotators assign the same label to the same items
Adjudication	Reviewing evidence and rules to resolve a disputed label or retain uncertainty

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Label Quality, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 6 • WORKED EVIDENCE

1. Two annotators label 20 leaves and agree on 16.
2. Observed agreement = 16 ÷ 20 = 80%.
3. Most disagreements concern tiny spots near the boundary.
4. Define a minimum spot rule, relabel a fixed subset, and report the new agreement.

Step 1

STEP 1 Two **annotator**s label 20 leaves and agree on 16.

Step 2

STEP 2 Observed **agreement** = 16 ÷ 20 = **80%**.

Step 3

STEP 3 Most dis**agreement**s concern tiny spots near the boundary.

Step 4

STEP 4 Define a minimum spot rule, relabel a fixed subset, and report the new **agreement**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Worked explanation · Label Quality

For B, the full lane is visible and a wheel touches its boundary. One **annotator** treated touching as clear; the other treated it as blocked. Rule v1 explicitly includes touching, so the reviewed label is blocked.

Preserve both original labels and record the item, rule version, **evidence**, decision and reviewer. Do not silently overwrite the originals.

For C, a curtain hides part of the lane. One **annotator** guessed clear; the other requested review. Rule v1 requires full visibility for clear. The justified status remains review until sufficient **evidence** is available.

More votes cannot reveal the hidden area.

Using rule v1, explain the decisions for B and C. If you exclude C from the original comparison, what is agreement among the remaining pairs? Why must you still report the exclusion?

A decision log could say: B, original clear/blocked, wheel touches line, v1, blocked after review by the teacher. For C it could say: original review/clear, lane partly hidden, v1, unresolved review.

Add the labeling method and rule version to the dataset card.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Label Quality

Write a **label rule** that resolves the visible boundary case.

The facts we will use

> **Supplied facts**
>
> Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/6

Label Quality

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Apply visible rule

X is spotted; Y is healthy under this exercise's definition.



2. Handle uncertainty

Z requires review; hidden tissue cannot be treated as visibly spot-free.



3. Review disagreement

If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

CHANGE ONE THING

Z receives a new fully visible image with no brown spots.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Apply visible rule	X is spotted; Y is healthy under this exercise’s definition.
2. Handle uncertainty	Z requires review; hidden tissue cannot be treated as visibly spot-free.
3. Review disagreement	If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

Keep the stated limit

Read the labelled picture: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Z receives a new fully visible image with no brown spots. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Label Quality claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model can learn the correct concept even when people label the same example differently. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE The model can learn the correct concept even when people label the same example differently.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Report the paired-item count, **agreement** count, handling of review and the rule version. For the original four-item comparison, say two of four pairs match, including review statuses.

If you report a subset, name its selection rule and denominator. This tiny invented set says nothing about reliability on other scenes.

Check the claim against the facts

Keep practice for writing **label rule**s separate from final model testing. Resolving labels needs a documented **process**; reviewing model **prediction**s to choose convenient labels would undermine a fair evaluation.

Next week you will separate related examples when designing data splits.

Two annotators agree on every item but both ignore the boundary rule. What does perfect agreement establish, and what does it fail to establish? Give one evidence check to perform next.

STUDENT LESSON · 8 OF 18

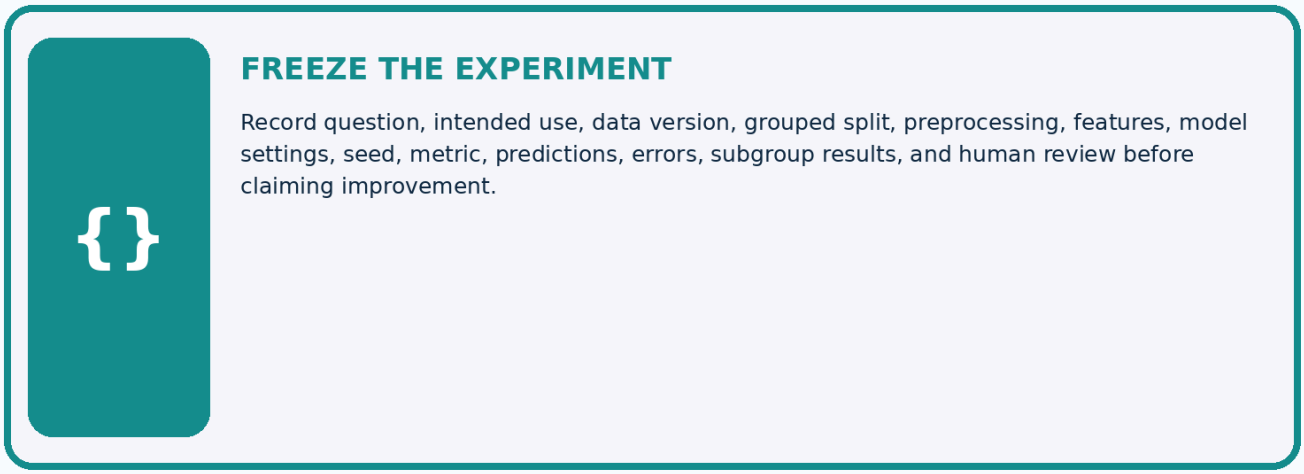
Practise together

Practise Label Quality with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A useful rule names the object of judgment, allowed labels, observable **evidence**, boundaries and an uncertainty route. Include a typical example and a boundary example.

Pilot the rule independently on a few new items, compare labels, discuss causes, revise the wording and label new items again. Record which rule produced each set of labels.

Do not change the rule simply to improve a score or make a model look correct. A rule revision can change the target itself.

If touching is changed from blocked to clear, document the changed task, review all affected examples consistently and keep versions separate.

Consult task **evidence** when interpreting a dis**agreement**; a majority vote alone cannot prove truth.

Practise with less help: Interpret agreement

Use workbook W2 for the connected example “Label definitions and dis**agreement**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Label definitions and dis**agreement**. Proposed explanation: “**Agreement** is accuracy”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should two **annotator**s not vote away Z’s missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

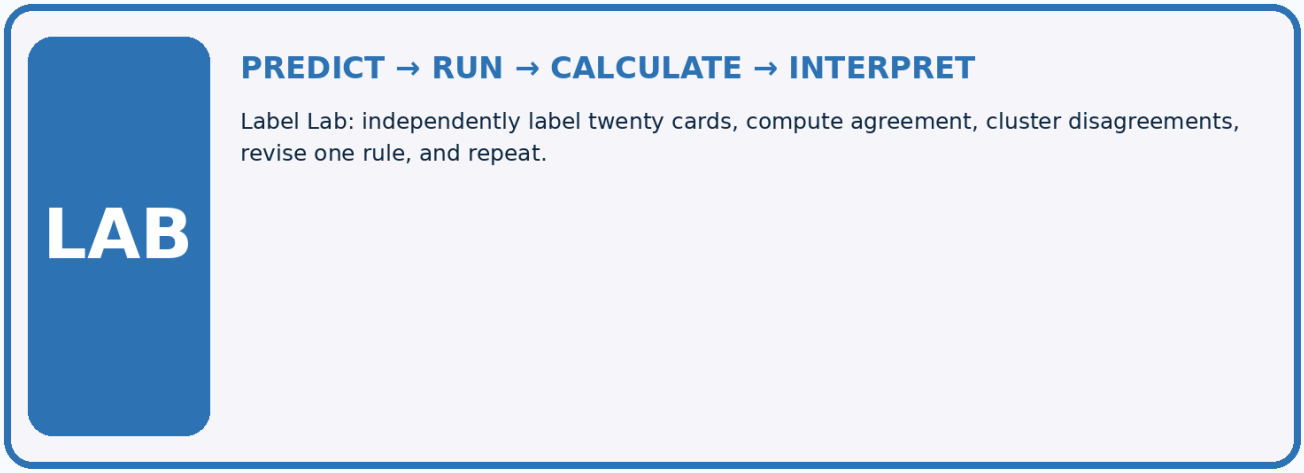
Run the investigation

> Investigate Label Quality. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Label Lab: independently label twenty cards, compute **agreement**, cluster dis**agreement**s, revise one rule, and repeat.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

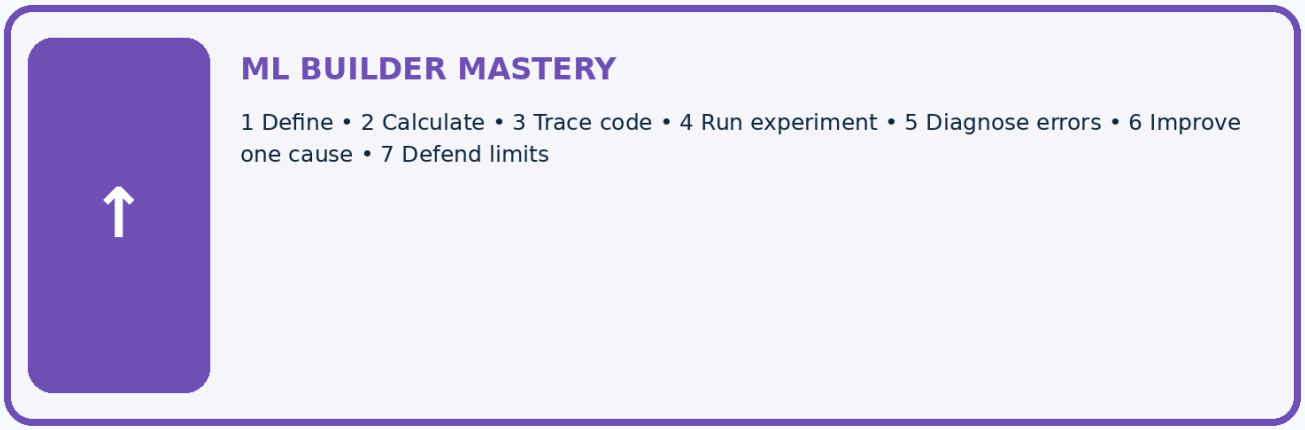
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Label Quality. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **annotation**, **label rule**, **agreement**, **adjudication**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The model can learn the correct concept even when people label the same example differently.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

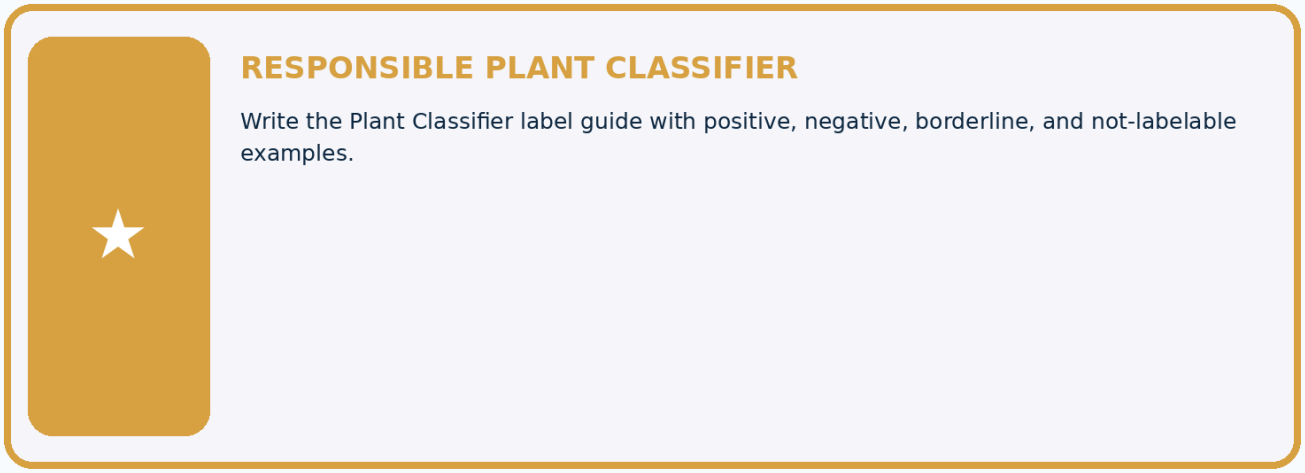
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



> THIS WEEK'S CHECKPOINT Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Measure **annotator agreement** and revise the labeling guide.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Sampling and Intended Use.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Label Quality case for supported practice. Repeated examples do not become fresh assessments.

Week 6 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

The question

Improve label quality.

Worked explanation

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Agreement is useful evidence, not truth

Evidence or stage	Value or result
Annotator agreement	7/10
Disagreement	3/10

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply the process independently

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 6 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better **process**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Label Quality, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Write a **label rule** that resolves the visible boundary case.

By the end, I can	How I will show it
Write an observable label rule with boundaries and a review option	Explain the deciding step, show a trace or test, and state one limit.
Measure agreement using matched items and an explicit denominator	Explain the deciding step, show a trace or test, and state one limit.
Investigate disagreement and record justified label decisions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Label Quality without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Write an observable **label rule** with boundaries and a review option

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a **label rule** that resolves the visible boundary case.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

1. Apply visible rule

2. Review dis**agreement**

3. Handle uncertainty

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review dis**agreement** If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Read the picture in order.

If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Z receives a new fully visible image with no brown spots. Which conclusion is supported by the supplied facts?

1. Label the new **observation** healthy under the toy rule.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Label the new **observation** healthy under the toy rule. The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Repair a missing building block

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

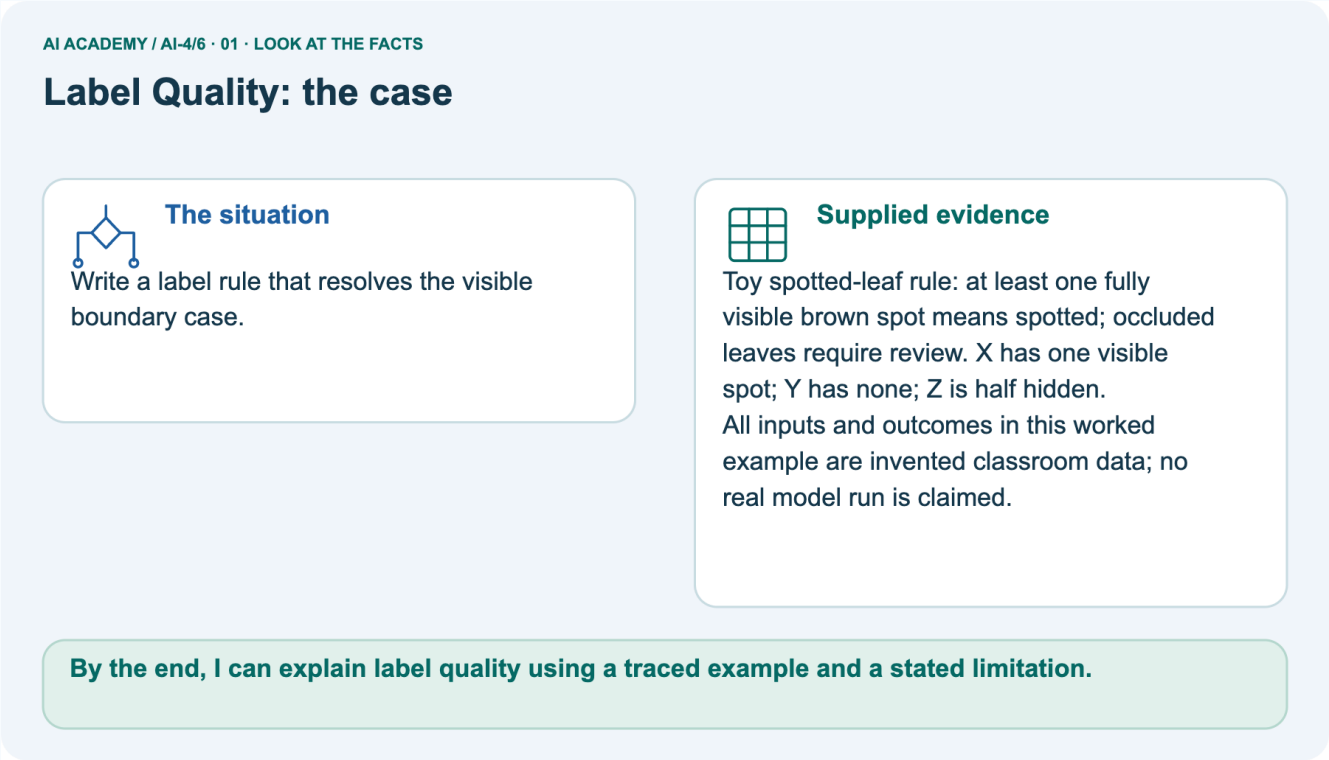
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Label Quality graphic using its labels, deciding evidence and relevant limitation.

01. Label Quality: the case



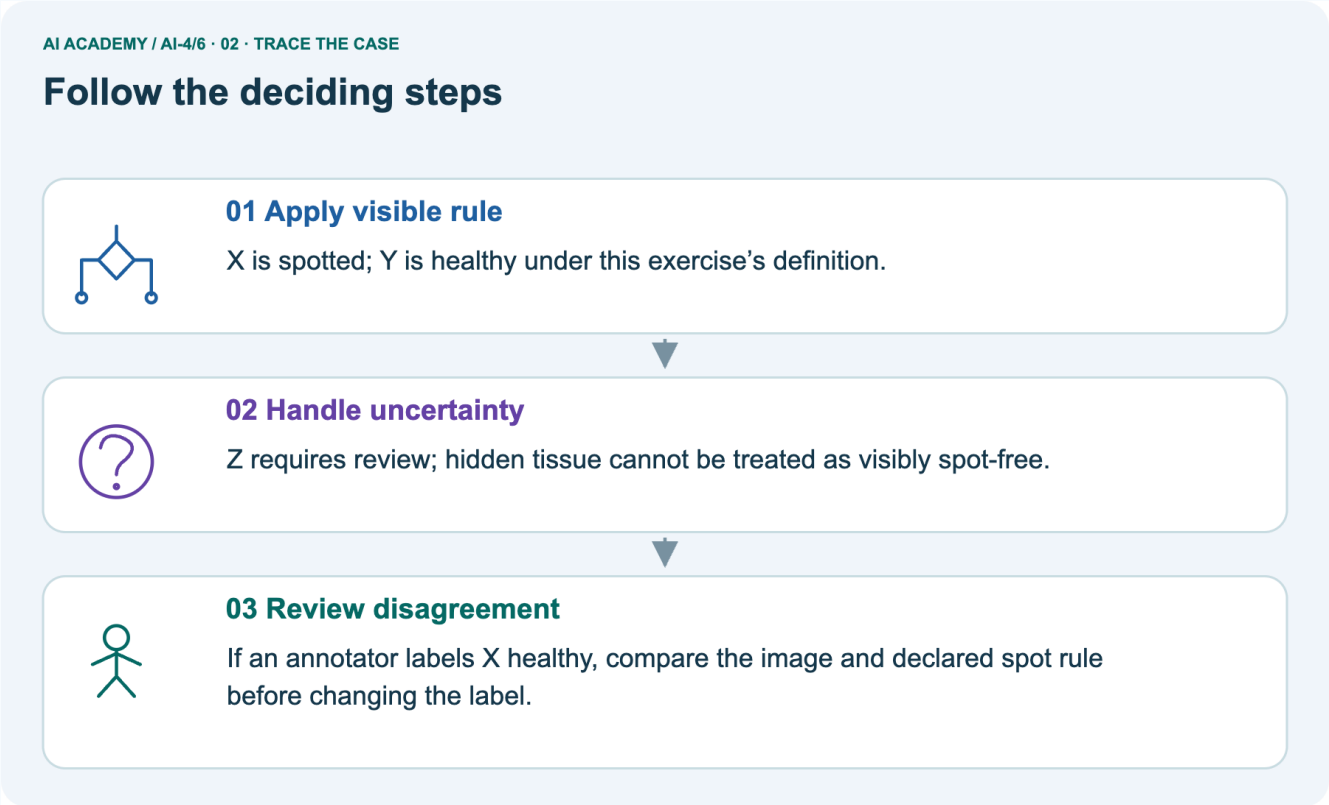
Follow the labelled picture with the matching explanation. Open the vector diagram

Write a **label rule** that resolves the visible boundary case. Supplied facts: Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review.

X has one visible spot; Y has none; Z is half hidden. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply visible rule: X is spotted; Y is healthy under this exercise's definition. → Handle uncertainty: Z requires review; hidden tissue cannot be treated as visibly spot-free. → Review dis**agreement**: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow a disagreement back to evidence



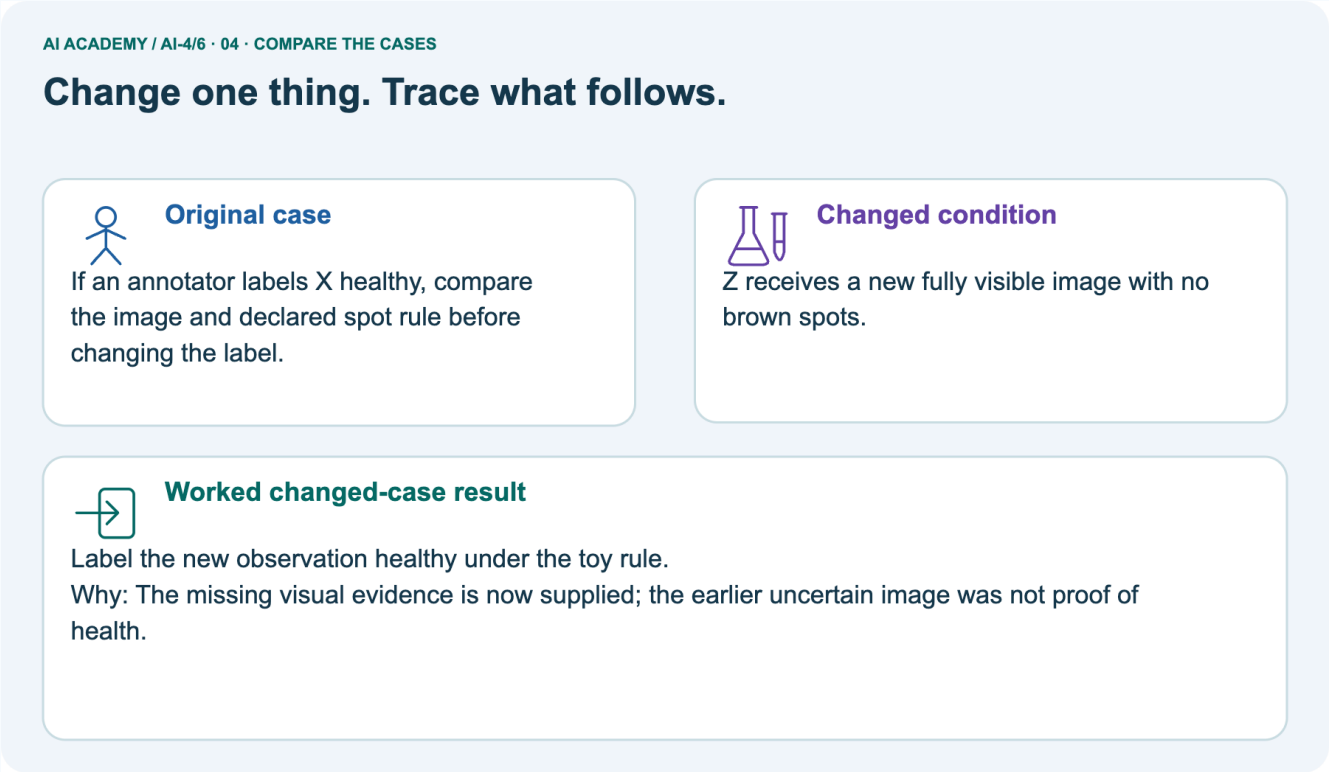
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow a dis**agreement** back to **evidence**. B has two labels | Read boundary rule | Keep labels and reason. C includes review | Check visible **evidence** | Retain uncertainty if needed.

A and D match | Check rule as well | **Agreement** is not proof

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



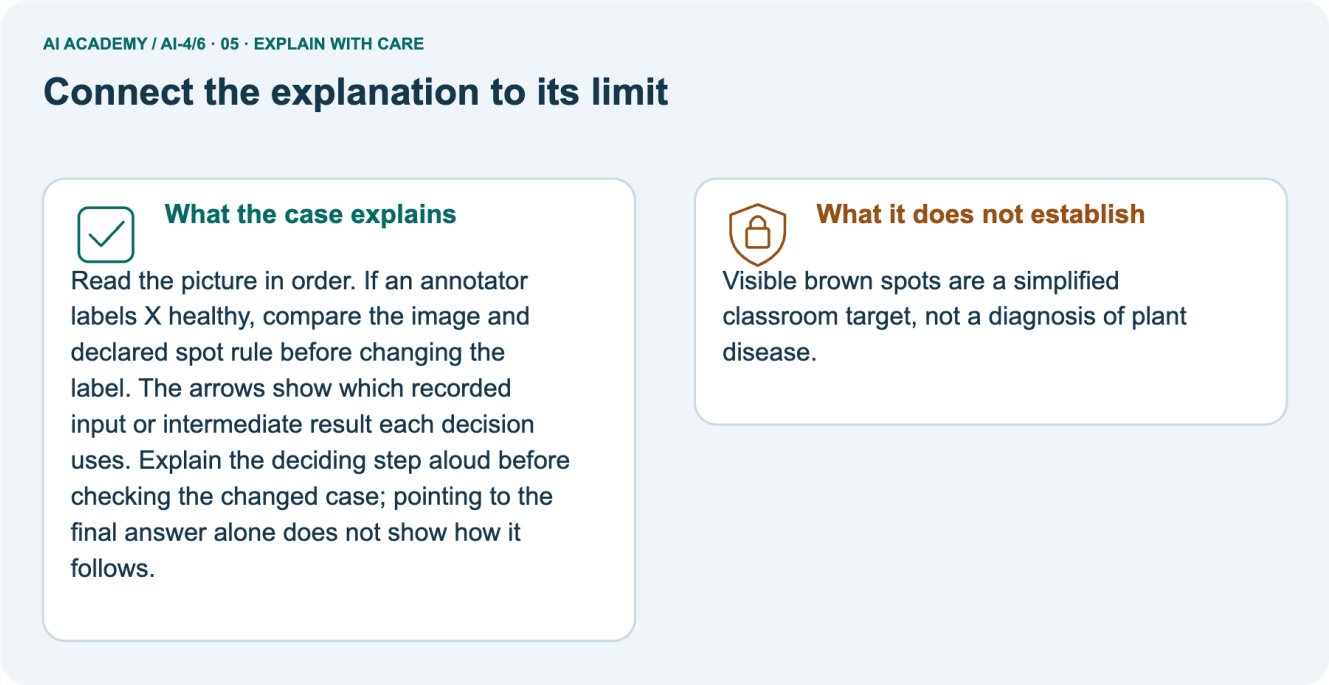
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Changed condition: Z receives a new fully visible image with no brown spots.

Expected result: Label the new **observation** healthy under the toy rule. Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



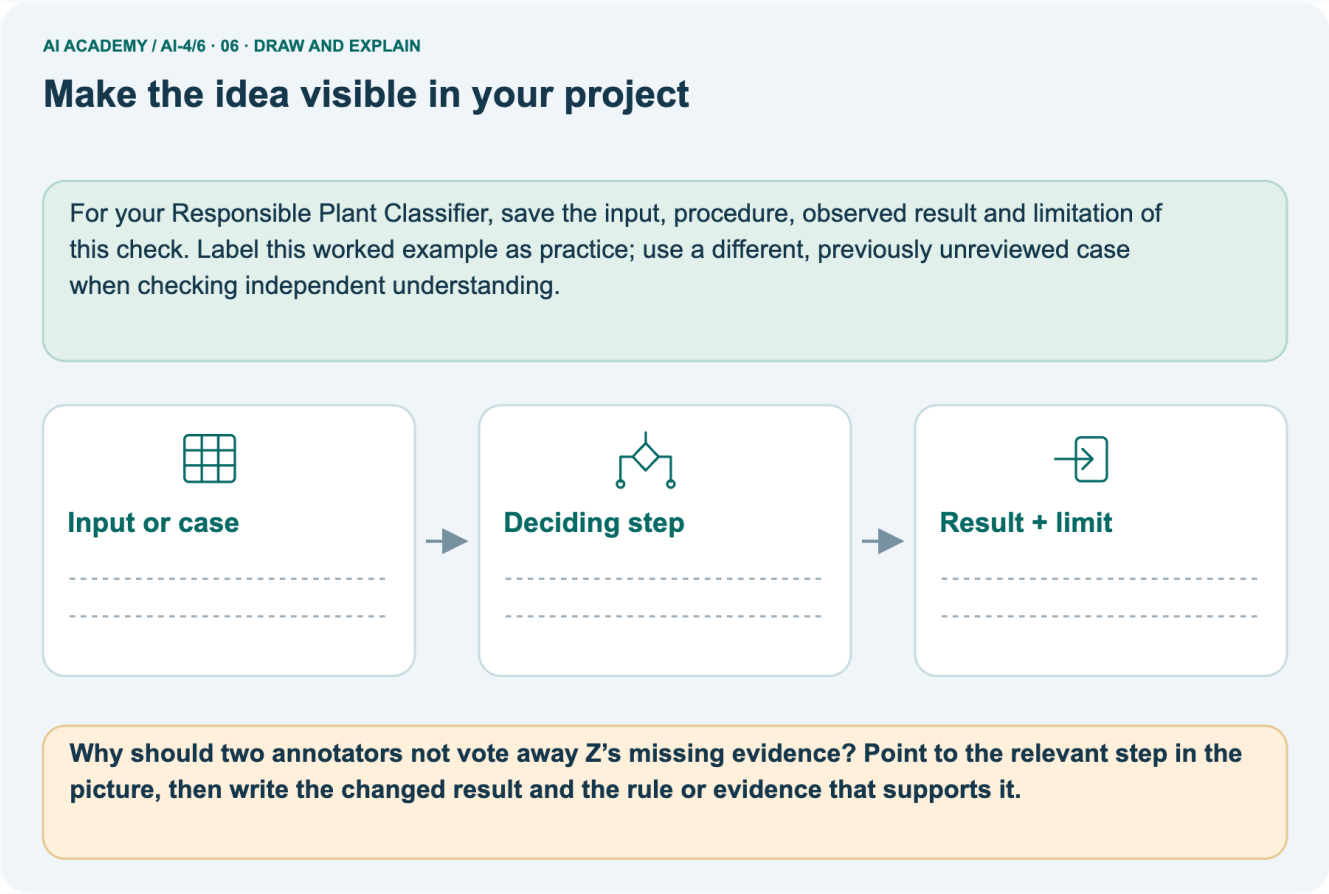
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.